

1 Product Overview

The portable integrated UAV detection and countermeasure device integrates detection, tracking, positioning, countermeasure and networking functions. A single device can complete the full UAV control workflow. With sufficient endurance, lightweight construction and reinforced design, it is well suited for backpack or handheld use.

The device scans UAV spectrum characteristics and decodes Drone ID and Remote ID messages to detect unauthorized UAV intrusions. It can identify control and video-transmission signals between UAVs and controllers, and uses electromagnetic suppression to interfere with UAV remote-control, video-transmission and navigation links, forcing UAVs to return or land.



Figure 1 Portable Integrated UAV Detection and Countermeasure Device

2 Key Features

- Integrated Detection and Countermeasure

Integrated detection and countermeasure design with high integration; one device can complete UAV detection and response.

- Broad UAV Model Detection

Supports detection and identification of consumer, industrial and non-standard UAVs in complex electromagnetic environments, with low false-alarm and missed-alarm rates.

- Long Countermeasure Distance

Up to 2 km in open environments (typical target: Mavic 2).

- Directional Search

Uses passive RF detection technology to provide 360° omnidirectional detection while stationary; after warning, it can switch to directional detection to determine the UAV direction.

- Accurate Identification

Accurately identifies the body serial number, model, position, speed, altitude, flight track and controller position of mainstream UAV brands.

- Precise Countermeasure

For some DJI UAV models, it can precisely counter one UAV among multiple UAVs while allowing the others to fly normally.

- Multiple Countermeasure Effects

Provides multi-band RF jamming capability to make UAVs return, land or hover.

● Portable Operation

Small, lightweight and easy to carry.

Basic Specifications	
Detection Mode	Supports both omnidirectional detection and directional direction finding
Multiple Algorithms Supported	Spectrum recognition, Drone ID, Remote ID
Detection Frequency Band	433M/800M/900M/ 1.4G/2.4G/5.2G/5.8G
Direction-Finding Accuracy	$\leq 30^\circ$ (directional detection); $\leq 0.5^\circ$ (protocol decoding)
Maximum Detection / Direction-Finding Distance	≥ 3 km (in open environment)
Countermeasure Frequency Band	800M/900M/ 1.2G/ 1.4G/ 1.6G/2.4G/5.2G/5.8G
Maximum Effective Jamming Distance	≥ 2 km (line-of-sight conditions)
Jamming-to-Control Range Ratio	$\geq 20:1$ (typical value: MAVIC 2)
Operating Time	Detection standby time ≥ 1.5 h; continuous countermeasure operation ≥ 15 min

Power Supply Method	Replaceable lithium battery
Operating Temperature	-20°C ~ +55°C
Product Form	Shield-shaped design
Networking Mode	Can connect to a unified UAV platform
Control Mode	10.1-inch touchscreen operation
Maximum Total Power Consumption	480W
Protection Rating	IP65
Weight	6.5KG